
EDUCATION

Ph.D., Aerospace Engineering Wichita State University (WSU)	Aug 2021 – May 2026 Kansas, USA
M.S., Aeronautical Engineering Linköping University	Sep 2017 - June 2019 Sweden
B.S., Aerospace Engineering Politehnica University of Bucharest	Oct 2013 - June 2017 Romania

RESEARCH EXPERIENCE

Wichita State University <i>Graduate Research Assistant</i>	Aug 2021 – May 2026 Wichita, Kansas
• Designed a new algorithm for low-thrust orbit-raising trajectory optimization to different target orbits: GEO, Molniya, NRHO. • Worked on a simulation framework for investigating the electrodynamic control of polymer fibers in electrospinning for in-space manufacturing. • Supported collaboration with computer science researchers from Kansas State University. • Developed system identification method for an air bearing platform for nanosatellite testing.	
National Institute for Aerospace Research "Elie Carafoli" – INCAS <i>Research Engineer Assistant</i>	Sep 2019 - Dec 2020 Bucharest, Romania
• Assembled finite element models for ongoing projects, e.g. Space Rider. • Created in-house computational tools for fastener analysis and technical report writing.	
Linköping University <i>Master's Researcher</i>	Jan 2019 - June 2019 Linköping, Sweden
• Implemented a system identification tool for subscale flight testing.	

AWARDS

- Received Dora Wallace Hodgson Outstanding Doctoral Student Award and Jayne & Glenn Milburn Fellow from Graduate School at WSU in 2025.
- Received the AFOSR Student Travel Award from the Space Flight Mechanics Committee of the American Astronautical Society at the 2025 AAS/AIAA Astrodynamics Specialist Conference.
- Achieved the 2nd place in the Proficient Category at the 2nd Annual Engineering Research Symposium, WSU, 2024.
- Received Erasmus scholarship to study as an exchange student at Istanbul Technical University, Turkey, Spring 2016.

ADDITIONAL EXPERIENCE

- Served as Teaching Assistant for Dynamics and Flight Control Systems courses, WSU (2025-2026).
- Mentored one undergraduate student in solving low-thrust trajectories with GMAT, a specialized software for spacecraft mission analysis, as part of an undergraduate research collaboration (NASA/WSU Jump Start Program, Spring 2025).

- Served as a student committee member on the Associate Dean of Research (ADR) search for the College of Engineering (2024).
- Assisted in Engineering Open House outreach activity. E.g., helped to present the Astronautics Lab (2022) and assisted visitors with the flight simulator in the General Aviation Flight Lab (2024).
- Served as a substitute instructor in the Modern Flight Control System Design I course (Spring 2024).
- Participated in the aero design competition Future Flight Design 2015, designed and built radio-controlled model aircraft.
- Competed in local and national level student contests in strength of materials, mechanics, mathematics in 2014 and 2015.
- Reviewed articles for Acta Astronautica, IEEE Control Systems Society Conference, Journal of Spacecraft and Rockets, IEEE Transactions on Aerospace and Electronic Systems, American Control Conference.

SKILLS

- Computer: MATLAB/Simulink, CATIA v5, ANSYS Fluent/CFX, MSC Nastran/Patran/Apex, Altair HyperMesh, Python, COMSOL Multiphysics.

PUBLICATIONS

Journal Articles

- S. M. T. Zaidi, **A. Arustei**, A. Munir and A. Dutta, *Single-Agent Attention Actor–Critic: A Deep Reinforcement Learning-Based Solution for Low-Thrust Spacecraft Trajectory Optimization*, in IEEE Transactions on Aerospace and Electronic Systems, vol. 61, no. 5, pp. 11628-11643, Oct. 2025, doi: 10.1109/TAES.2025.3569615
- S. M. T. Zaidi, **A. Arustei**, A. Munir and A. Dutta, *Automated Trajectory Planning: A Cascaded Deep Reinforcement Learning Approach for Low-Thrust Spacecraft Orbit-Raising*, IEEE Aerospace and Electronic Systems Magazine, Jan 2025. doi: 10.1109/MAES.2025.3556795.
- **A. Arustei**, A. Dutta, *Direct Optimization of Low-thrust Orbit-raising Maneuvers Using Adjoint Sensitivities*, Acta Astronautica, Vol. 219, 2024, pp. 965–981, doi: 10.1016/j.actaastro.2024.03.059.

Conference Papers

- J. Steck, **A. Arustei**, A. Dutta. *Low Thrust Minimum Time Transfer: Reinforcement Learning of Optimal Co-State Time Dependence*, AIAA 2026-2782, AIAA SCITECH 2026 Forum, January 2026. (in press)
- **A. Arustei**, A. Dutta, *Nested B-spline Control and Modified (h,e) Elements for Lengthy Low-thrust Cislunar Orbit-raising*, in: AAS/AIAA Astrodynamics Specialist Conference, Boston, Massachusetts, USA, 2025, AAS 25-874.
- James E. Steck, **A. Arustei**, Atri Dutta, *Low Thrust Minimum Time Transfer: Single Network Adaptive Critic Approximate Dynamic Programming*, 2025 AAS/AIAA Space Flight Mechanics Meeting, AAS 2025-311 (in press).
- Atri Dutta, **A. Arustei**, Matthew Chace, Pardhasai Chadavada, James Steck, Talha Zaidi, Arslan Munir, *Machine Learning Assisted Low-Thrust Orbit-Raising: A Comparative Assessment of a Sequential Algorithm and Deep Reinforcement Learning Approach*, AIAA SCITECH 2024 Forum, Jan 2024, doi: 10.2514/6.2024-1669.
- **A. Arustei**, A. Dutta, *An adjoint sensitivity method for the sequential low-thrust orbit raising problem*, in: AAS/AIAA Astrodynamics Specialist Conference, Charlotte, North Carolina, USA, 2022, AAS 22-835.

Theses/Dissertations

- **A. Arustei**, *Computational Graph Construction of Adjoint Sensitivities for Low-thrust Trajectory Optimization and Desensitized Optimal Control*, Wichita State University, Doctoral Dissertation, 2026.
 - **A. Arustei**, *Development of a System Identification Tool for Subscale Flight Testing*, Linköping University, Master's Thesis, 2019, DiVA id: diva2:1324418.
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